Squares, Cubes and Roots

Concept

Squares and cubes describe repeated multiplication. Roots undo those powers.

Perfect square

$n^2 = n \times n$; perfect cube: $n^3 = n \times n \times n$.

Principal root

$\sqrt{a}$ is non-negative. $\sqrt[3]{a}$ can be negative.

Estimate

Place a number between two nearby perfect squares.

Watch the sign

$\sqrt{49} = 7$, but $x^2 = 49$ has solutions ± 7.

Example 1

Find the side and perimeter of a square with area $196\ m^2$.

Solution

1. Side $= \sqrt{196} = 14$ m.

2. Perimeter $= 4 \times 14 = 56$ m.

Squares, Cubes and Roots

Example 2

Is 180 a perfect square? Estimate its square root.

Solution

1. $180 = 2^2 \times 3^2 \times 5$. The factor 5 is unpaired: not a perfect square.

2. $169 < 180 < 196$, so $13 < \sqrt{180} < 14$.

Squares, Cubes and Roots

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find $\sqrt{225}$, $\sqrt[3]{512}$ and $\sqrt[3]{(-125)}$.

2. Find $\sqrt{(25/144)}$.

3. Between which whole numbers is $\sqrt{85}$?

4. A square has perimeter 52 cm. Find its area.

5. Is 432 a perfect cube? Explain using prime factors.

Squares, Cubes and Roots

Practice answers | check after trying

1. 15, 8 and -5; squaring or cubing returns the original values.

2. 5/12.

3. Between 9 and 10 because $81 < 85 < 100$.

4. Side = 52/4 = 13 cm; area = $13^2 = 169$ cm^2.

5. No. $432 = 2^4 \times 3^3$, so the four factors of 2 cannot all form groups of three.

Fraction Operations

Concept

Fractions name equal parts of a whole. Choose the operation before calculating.

Add / subtract

Use a common denominator; combine only numerators.

Multiply

Multiply numerators and denominators; cancel common factors.

Divide

Multiply by the reciprocal of the non-zero divisor.

Order

Brackets → powers → multiply/divide → add/subtract.

Example 1

Calculate $-2/3 + 5/8$.

Solution

1. Common denominator: 24.

2. $-2/3 + 5/8 = -16/24 + 15/24 = -1/24$.

Fraction Operations

Example 2

Calculate $1\ 1/2 \div 2\ 1/4$.

Solution

1. Convert mixed numbers: $3/2 \div 9/4$.

2. Multiply by reciprocal: $(3/2)(4/9) = 2/3$.

Example 3

Calculate $3/4 - (1/2)(2/3)$.

Solution

1. Multiply first: $(1/2)(2/3) = 1/3$.

2. Then subtract: $3/4 - 1/3 = 9/12 - 4/12 = 5/12$.

Fraction Operations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. $7/10 + 1/6$.

2. $-3/4 - 5/8$.

3. $(4/9)(-3/8)$.

4. $5/6 \div (-10/9)$.

5. $2 - (3/4)^2$.

6. A 3 1/2 m ribbon is cut into 1/4 m lengths. How many full pieces?

Fraction Operations

Practice answers | check after trying

1. 21/30 + 5/30 = 26/30 = 13/15.

2. -6/8 - 5/8 = -11/8.

3. -12/72 = -1/6.

4. (5/6)(-9/10) = -3/4.

5. 32/16 - 9/16 = 23/16.

6. (7/2) ÷ (1/4) = 14 pieces. Check: 14 × 1/4 = 3 1/2 m.

Ratios, Rates and Proportions

Concept

Ratios compare quantities. Proportions describe equal ratios.

Ratio order

Red:blue is different from red:total. Match units first.

Unit rate

Divide to find the amount per 1 unit.

Proportion

$a/b = c/d$ means $ad = bc$, with non-zero denominators.

Sharing

Add ratio parts → find one part → multiply each share.

Example 1

Share 96 stickers in the ratio 3:5.

Solution

1. Total parts $= 3 + 5 = 8$. One part $= 96/8 = 12$.

2. Shares $= 3(12) = 36$ and $5(12) = 60$ stickers.

Ratios, Rates and Proportions

Example 2

Six notebooks cost $15. Find the cost of nine at the same unit price.

Solution

1. Unit price = 15/6 = $2.50 per notebook.

2. Nine notebooks cost 9 × 2.50 = $22.50.

Ratios, Rates and Proportions

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Simplify 24:36:60.

2. Find the speed for 150 km in 2.5 h.

3. Share $84 in the ratio 2:5.

4. Solve $x/8 = 15/12$.

5. A delivery costs $6 plus $2 per km. Is cost proportional to distance? Explain.

Ratios, Rates and Proportions

Practice answers | check after trying

1. Divide by 12: 2:3:5.

2. 150/2.5 = 60 km/h.

3. One part = 84/7 = 12; shares $24 and $60.

4. 12x = 120, so x = 10.

5. No. At zero distance the fee is $6, and the cost per kilometre changes with distance.

Percents and Percent Change

Concept

Percent means per hundred. Always identify the reference whole.

Convert

Percent ÷ 100 = decimal; 0.4% = 0.004.

Find a part

Part = decimal rate × whole.

Change

Percent change = (new − original)/original × 100%.

Reverse

New = original × (1 ± rate); divide to recover original.

Example 1

Find 0.4% of ₹75,000.

Solution

1. 0.4% = 0.004.

2. Part = 0.004 × ₹75,000 = 300.

Percents and Percent Change

Example 2

A price is $54 after a 25% discount. Find its original price.

Solution

1. Remaining fraction = 1 − 0.25 = 0.75.
2. Original = 54/0.75 = $72.
3. Check: 72 × 0.75 = 54.

Example 3

Find the net change after a 20% rise and a 20% fall.

Solution

1. Combined multiplier = 1.20 × 0.80 = 0.96.
2. The final amount is 96% of the original: a 4% decrease.

Percents and Percent Change

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Convert 0.08% to a decimal.

2. Find 240% of 35.

3. 18 is what percent of 48?

4. A value falls from 160 to 124. Find its percent decrease.

5. A price is $69 after a 15% increase. Find the original.

6. Find the net change after a 10% discount followed by another 10% discount.

Percents and Percent Change

Practice answers | check after trying

1. 0.0008.

2. 2.4 × 35 = 84.

3. 18/48 = 0.375 = 37.5%.

4. Decrease = 36; 36/160 = 22.5%.

5. 69/1.15 = $60.

6. 0.90 × 0.90 = 0.81, so the overall discount is 19%, not 20%.

Expressions and Two-Step Equations

Concept

An expression has a value. An equation states that two expressions are equal.

Substitute

Replace variables using brackets: $x = -3 \to 2(-3)^2$.

Like terms

Combine matching variable parts: $3x + 5x = 8x$.

Balance

Do the same operation to both sides of an equation.

Two steps

In $ax + b = c$, undo b, then divide by $a \neq 0$.

Example 1

Solve $4x - 9 = 19$.

Solution

1. Add 9 to both sides: $4x = 28$.
2. Divide by 4: $x = 7$.
3. Check: $4(7) - 9 = 19$.

Expressions and Two-Step Equations

Example 2

A rental costs $8 plus $6 per hour. Total: $38. Find the hours.

Solution

1. Let h = hours. Write $6h + 8 = 38$.

2. $6h = 30 \rightarrow h = 5$ hours.

3. Check: $6(5) + 8 = \$38$.

Expressions and Two-Step Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Evaluate $2p^2 + q$ for $p = -4$ and $q = 3$.

2. Solve $5x + 6 = 41$.

3. Solve $-3x + 4 = 22$.

4. Solve $x/4 - 7 = -2$.

5. Three identical tickets and a $4 fee cost $31. Find the price per ticket.

Expressions and Two-Step Equations

Practice answers | check after trying

1. $2(16) + 3 = 35$.

2. $5x = 35$, so $x = 7$.

3. $-3x = 18$, so $x = -6$.

4. $x/4 = 5$, so $x = 20$.

5. Let t be ticket price. $3t + 4 = 31$ gives $t = 9$. Each ticket costs $9; $3(9) + 4 = 31$.

Discrete Linear Patterns

Concept

A discrete linear pattern changes by the same amount at each whole-number stage.

Common difference

Subtract consecutive terms to find d.

Term rule

$T = a + (n - 1)d$; a is the first term, n starts at 1.

Discrete graph

Plot separate points when half-stages have no meaning.

Membership

Solve for n; accept only allowed whole-number stages.

Example 1

Find a rule and term 20 for $7, 11, 15, 19, \ldots$.

Solution

1. $d = 4$, so $T = 7 + 4(n - 1) = 4n + 3$.

2. At $n = 20$: $T = 4(20) + 3 = 83$.

Discrete Linear Patterns

Example 2

Is 100 a term of $T = 4n + 3$?

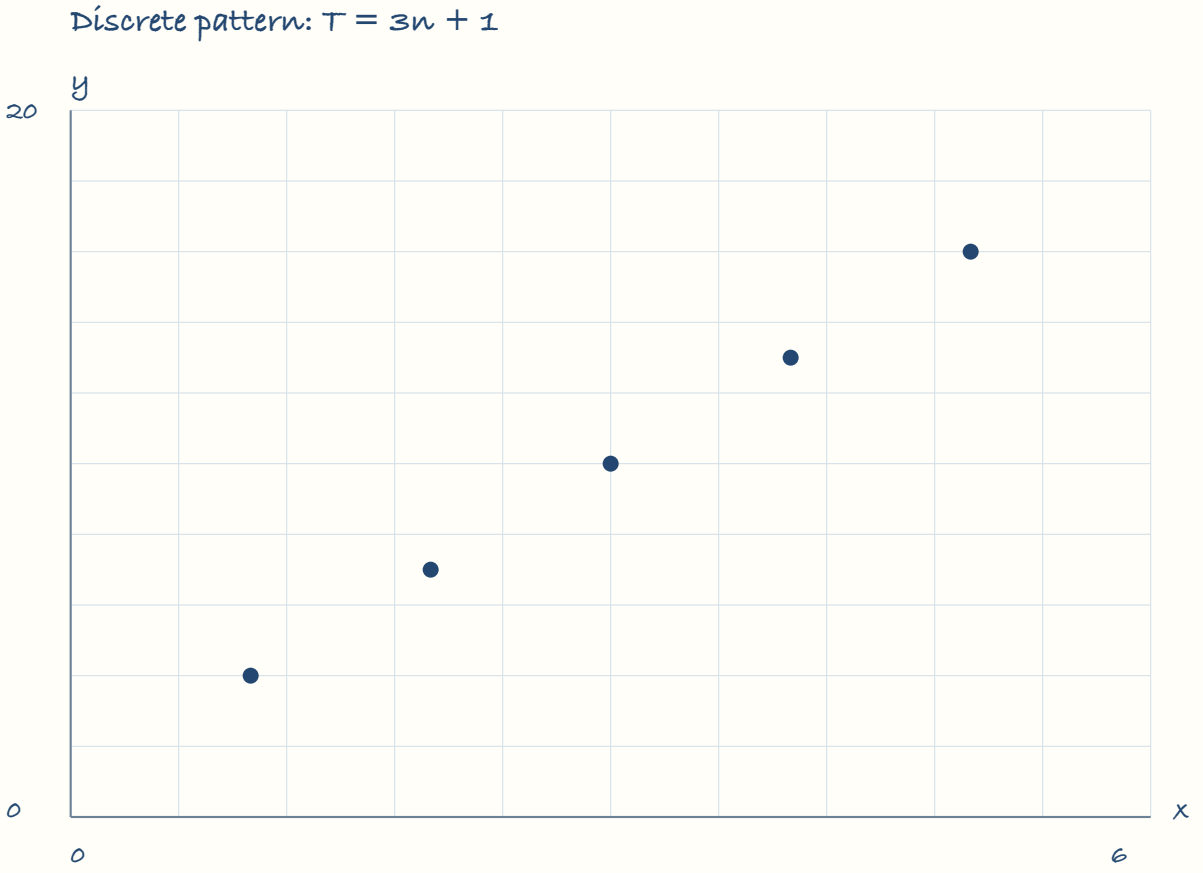
Solution

1. $100 = 4n + 3 \rightarrow n = 97/4 = 24.25$.

2. No: the stage number is not a whole number.

Discrete Linear Patterns

Visual notes | read the graph



Each dot is a whole-number stage. There is no half-stage in this model.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Discrete Linear Patterns

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Write a rule for 5, 8, 11, 14, … .

2. Find its 30th term.

3. For $T = 6n + 2$, list the first four terms.

4. Is 74 a term of $T = 6n + 2$?

5. Explain why $y = n^2$ does not describe a linear pattern.

Discrete Linear Patterns

Practice answers | check after trying

1. $T = 3n + 2$.

2. $3(30) + 2 = 92$.

3. 8, 14, 20, 26.

4. $6n + 2 = 74$ gives $n = 12$, so yes.

5. Outputs 1, 4, 9, 16 have differences 3, 5, 7. The difference is not constant.

The Pythagorean Theorem

Concept

The Pythagorean theorem connects the three sides of a right triangle.

Hypotenuse

c is opposite the 90° angle and is the longest side.

Formula

$a^2 + b^2 = c^2$, for a right triangle only.

Missing leg

$a = \sqrt{(c^2 - b^2)}$; subtract, then take the positive root.

Check

Use the same units; the hypotenuse must be longest.

Example 1

Find the hypotenuse when the legs are 9 cm and 12 cm.

Solution

1. $c^2 = 9^2 + 12^2 = 81 + 144 = 225$.

2. $c = \sqrt{225} = 15$ cm.

The Pythagorean Theorem

Example 2

A 13 m ladder reaches 12 m up a wall. Find the distance from the wall.

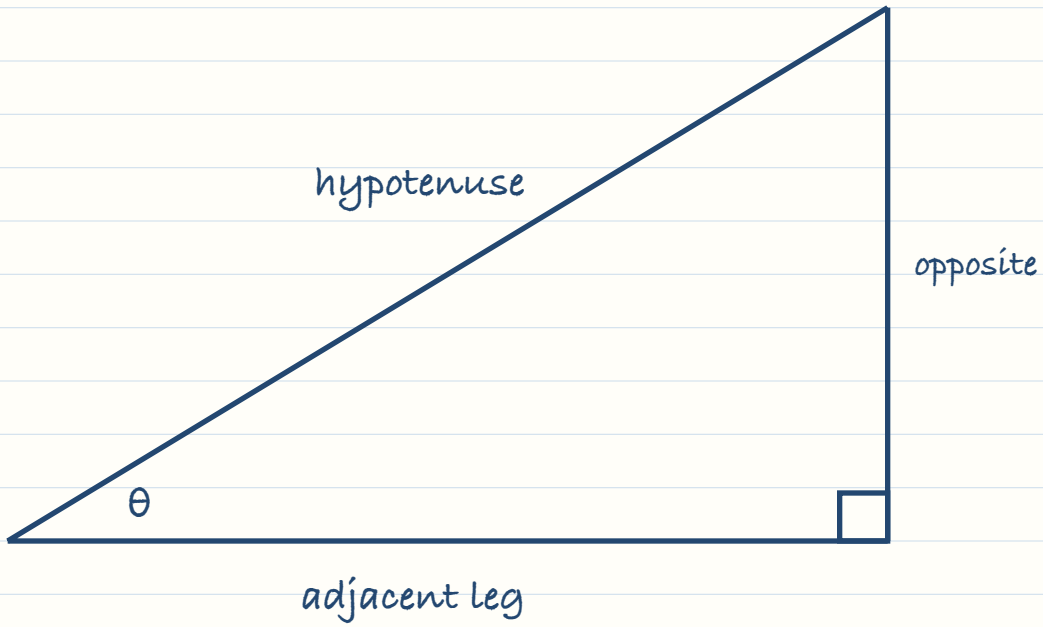
Solution

1. The ladder is the hypotenuse: $x^2 + 12^2 = 13^2$.

2. $x^2 = 169 - 144 = 25 \rightarrow x = 5$ m.

The Pythagorean Theorem

Visual notes | label before calculating



The hypotenuse is opposite the right angle. Opposite and adjacent refer to the selected acute angle θ. A different selected angle changes those two labels.

Here a and b label the two legs and c labels the hypotenuse. Squares built on the legs have total area $a^2 + b^2$, equal to c^2, the area of the square built on the hypotenuse.

The Pythagorean Theorem

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find the hypotenuse with legs 5 and 12.

2. Find a leg if the hypotenuse is 10 and the other leg is 6.

3. Is a triangle with sides 8, 15 and 17 right?

4. Find the diagonal of a square with side 7 cm, exactly and to two decimals.

The Pythagorean Theorem

Practice answers | check after trying

1. $\sqrt{(25 + 144)} = 13$.

2. $\sqrt{(100 - 36)} = 8$.

3. Yes: $64 + 225 = 289 = 17^2$.

4. $\sqrt{98} = 7\sqrt{2}$ cm ≈ 9.90 cm. The exact radical avoids rounding error.

Solids, Nets, Surface Area and Volume

Concept

Surface area measures the outside covering; volume measures space inside.

Right prism

$V = Bh$; $SA = 2B + Ph$. B = base area, P = base perimeter.

Cylinder

$V = \pi r^2 h$; closed $SA = 2\pi r^2 + 2\pi rh$.

Nets and views

Unfold all faces; top/front/side views may hide cubes.

Units

Area uses square units; volume uses cubic units. Remove missing faces.

Example 1

Find volume and surface area of a $6 \times 4 \times 3$ cm box.

Solution

1. $V = 6 \times 4 \times 3 = 72\ \text{cm}^3$.

2. $SA = 2(6\times4 + 6\times3 + 4\times3) = 108\ \text{cm}^2$.

Solids, Nets, Surface Area and Volume

Example 2

A closed cylinder has radius 3 cm and height 8 cm. Find V and SA.

Solution

1. $V = \pi(3^2)(8) = 72\pi \text{ cm}^3 \approx 226.19 \text{ cm}^3$.

2. $SA = 2\pi(3^2) + 2\pi(3)(8) = 66\pi \text{ cm}^2 \approx 207.35 \text{ cm}^2$.

Example 3

A triangular prism has a 3-4-5 right-triangle base and length 10 cm.

Solution

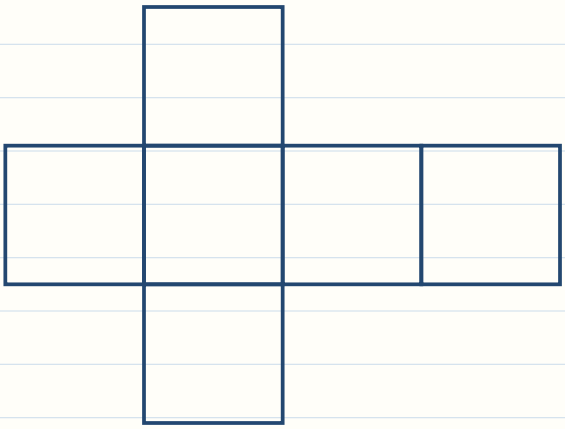
1. Base area $B = (3 \times 4)/2 = 6 \text{ cm}^2$; perimeter $P = 3+4+5 = 12$ cm.

2. $V = 6(10) = 60 \text{ cm}^3$; $SA = 2(6) + 12(10) = 132 \text{ cm}^2$.

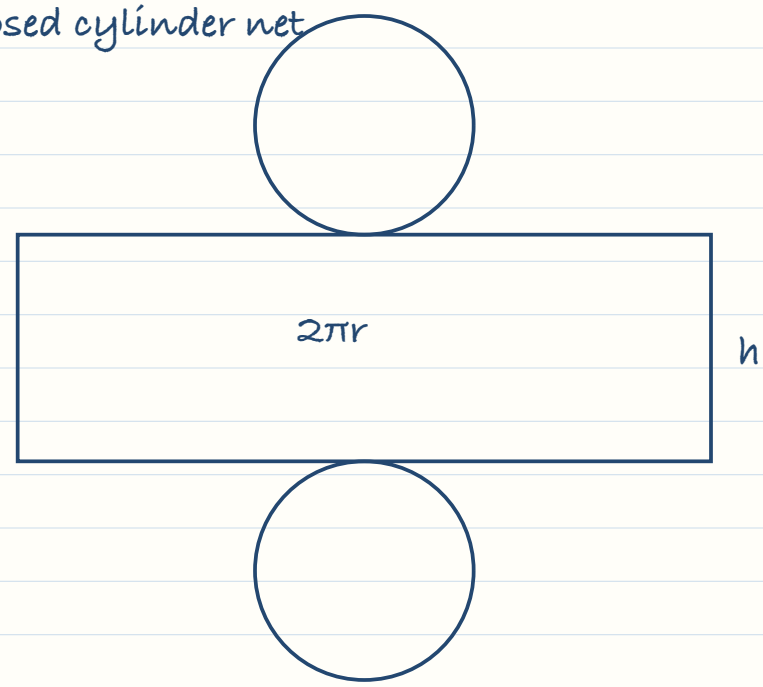
Solids, Nets, Surface Area and Volume

Visual notes | nets

A cube net: six equal squares



A closed cylinder net



Cylinder: two circles and a 2πr by h rectangle.

Sketches are schematic; match shared edges when folding.

Solids, Nets, Surface Area and Volume

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find V and SA of a cube with side 5 cm.

2. Find V of a cylinder with diameter 10 cm and height 6 cm.

3. An open box has length 8, width 5 and height 2 cm. Find its surface area.

4. What rectangle is needed for the curved surface of a cylinder of radius 2 cm and height 7 cm?

Solids, Nets, Surface Area and Volume

Practice answers | check after trying

1. $V = 125\ \text{cm}^3$; $SA = 6 \times 25 = 150\ \text{cm}^2$.

2. Radius $= 5$, so $V = 150\pi\ \text{cm}^3 \approx 471.24\ \text{cm}^3$.

3. Base + four walls $= 40 + 2(16) + 2(10) = 92\ \text{cm}^2$.

4. A rectangle 4π cm by 7 cm. Its length wraps exactly once around the circle.

Averages and Independent Probability

Concept

Averages describe a typical value. Probability describes how likely an event is.

Centre

Mean = sum/count; median = sorted middle; mode = most frequent.

Outlier

An extreme value affects the mean more than the median.

Probability

Favourable/total outcomes, when outcomes are equally likely.

Independent

P(A and B) = P(A)P(B); P(not A) = 1 - P(A).

Example 1

Find mean, median and mode of 2, 3, 3, 4, 18.

Solution

1. Mean = 30/5 = 6. Median = 3. Mode = 3.

2. The value 18 pulls the mean up; the median better represents a typical value here.

Averages and Independent Probability

Example 2

Toss two fair coins. Find the probability of exactly one head.

Solution

1. Sample space: HH, HT, TH, TT.

2. Successes: HT and TH → $P = 2/4 = 1/2$.

Averages and Independent Probability

Visual notes | two-stage sample space

	H	T
H	HH	HT
T	TH	TT

Rows show the first fair coin; columns show the second. Four equally likely ordered outcomes remain after the labels are removed. Exactly one head occurs in HT or TH, so its probability is 2/4 = 1/2.

A tree gives the same result: each first toss branches into H or T, and each of those branches splits into H or T again. Multiply probabilities along a path; add probabilities of disjoint successful paths.

Averages and Independent Probability

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find mean, median and mode for 4, 5, 5, 6, 10.

2. Toss two fair coins. Find P(at least one head).

3. Roll two fair dice. Find P(both show 6).

4. Why should a survey report more than just its mean?

5. Are two card draws without replacement independent?

Averages and Independent Probability

Practice answers | check after trying

1. Mean = 30/5 = 6; median = 5; mode = 5.

2. 1 - P(TT) = 1 - 1/4 = 3/4.

3. (1/6)(1/6) = 1/36.

4. Different distributions and outliers can have the same mean; sample size, spread and context help interpret it.

5. Generally no: removing the first card changes the possible outcomes and probabilities for the second draw.

Best Buys and Consumer Decisions

Concept

A best buy compares the cost of equal quantities and the amount you will use.

Unit price

Price ÷ quantity; compare in matching units.

Discount

Sale price = original × (1 − discount rate).

Final cost

Apply stated coupons, taxes and fees in the stated order.

Value

Consider waste, storage, quality and your budget too.

Example 1

Compare 750 g for $4.50 with 1.2 kg for $6.60.

Solution

1. First: 4.50/0.75 = $6/kg.
2. Second: 6.60/1.2 = $5.50/kg → cheaper per kilogram.

Best Buys and Consumer Decisions

Example 2

A $60 jacket is 20% off, then subject to a stated 5% classroom tax.

Solution

1. Sale price $= 60(0.80) = \$48.$

2. Total $= 48(1.05) = \$50.40.$

Best Buys and Consumer Decisions

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Which is cheaper per litre: 500 mL for $2.40 or 1.5 L for $6.30?

2. Find the total for an $80 item after a 25% discount and then a stated 10% tax.

3. Three identical items cost $9 with a 'buy two, get one free' offer at $4.50 each. Find the effective unit price.

4. For $40, apply 20% off then a $5 coupon. Compare with applying the coupon first and then 20% off.

Best Buys and Consumer Decisions

Practice answers | check after trying

1. $4.80/L versus $4.20/L; the 1.5 L package has the lower unit price.

2. $80(0.75)(1.10) = \$66$.

3. $9/3 = \$3$ per item.

4. Discount then coupon: $40(0.80) - 5 = \$27$. Coupon then discount: $(40 - 5)(0.80) = \$28$. The first order saves $1 more, if permitted.